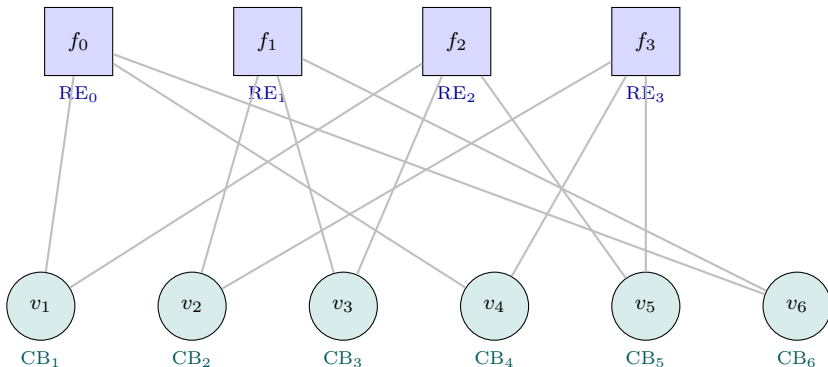


SCMA Factor Graph $\mathbf{F}_{4 \times 6}$

4 Resource Elements $\times$ 6 Codebooks



$d_f = 3$ (3 CBs per RE)

Overloading: $\frac{J}{K} = \frac{6}{4} = 150\%$

$d_v = 2$ (2 REs per CB)

Indicator Matrix:

$$\mathbf{F} = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$

Nominal (1 UE per CB): $d_c = d_f = 3$ at every RE.

Contention (CB_2 has 2 distinct DMRS combos):

$\text{RE}_1 = \{\text{CB}_2(2), \text{CB}_3(1), \text{CB}_6(1)\} \rightarrow d_c = 4$

$\text{RE}_3 = \{\text{CB}_2(2), \text{CB}_4(1), \text{CB}_5(1)\} \rightarrow d_c = 4$

MPA degrades: $p_{\text{BD}}(3) = 0.95$, $p_{\text{BD}}(4) = 0.85$